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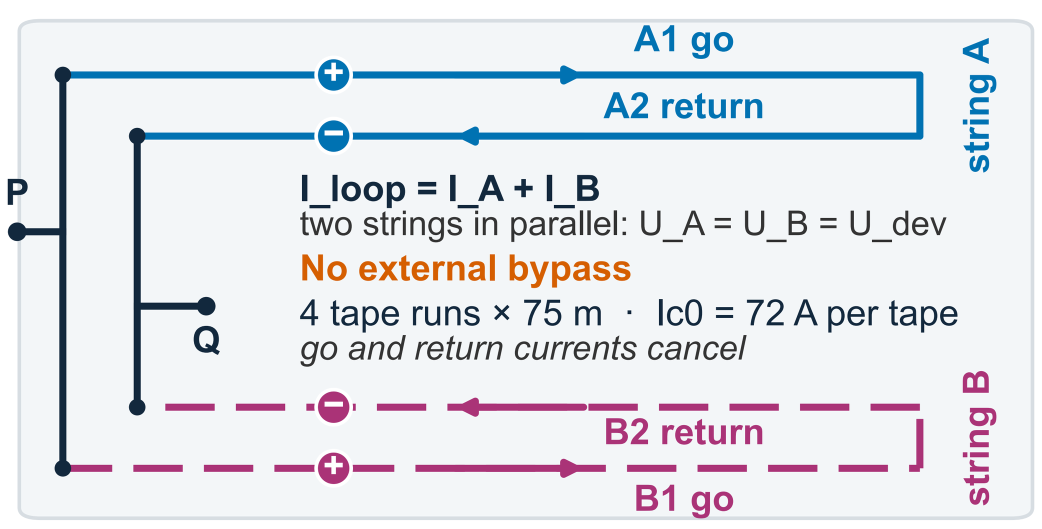
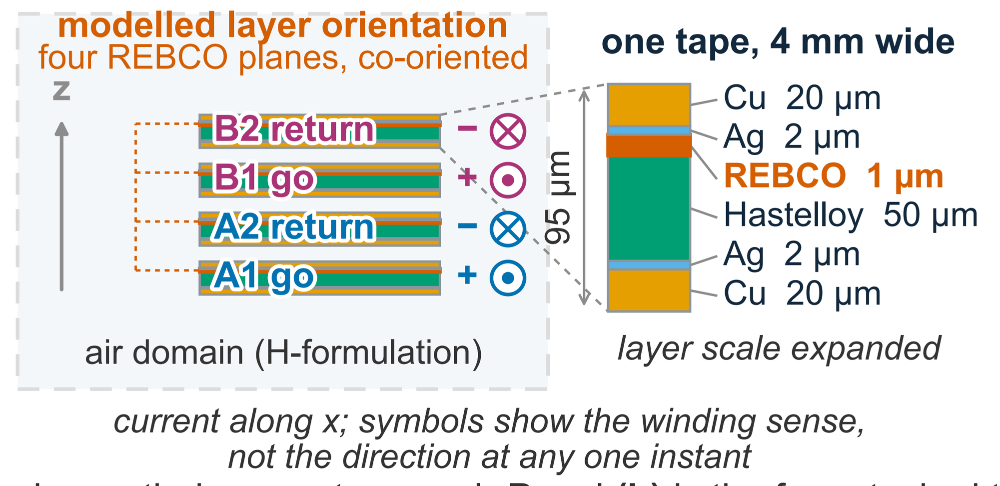
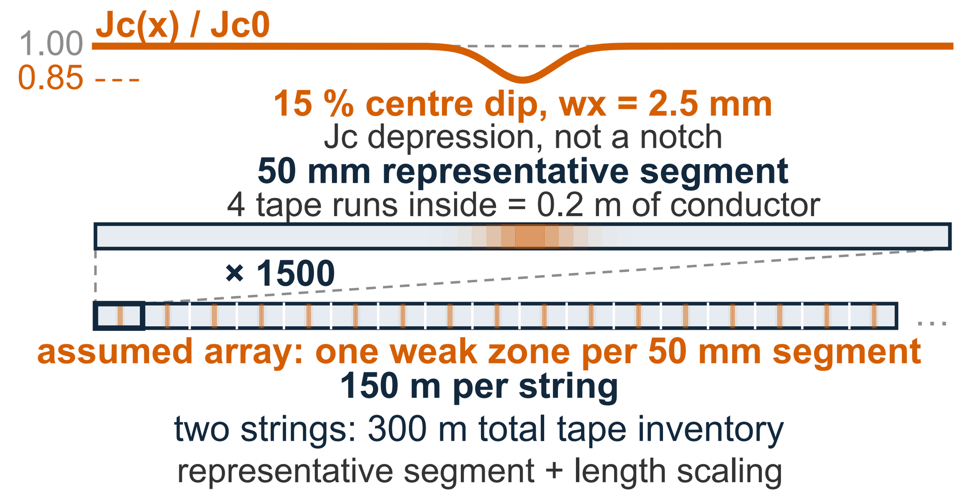
3D Coupled Modelling of a Series-Connected Bifilar Cu-Stabilised REBCO Resistive SFCL

H-formulation, solid heat transfer and external circuit solved together on a 50 mm representative segment of four tape runs, 400 V source model

1 Device topology and length scaling

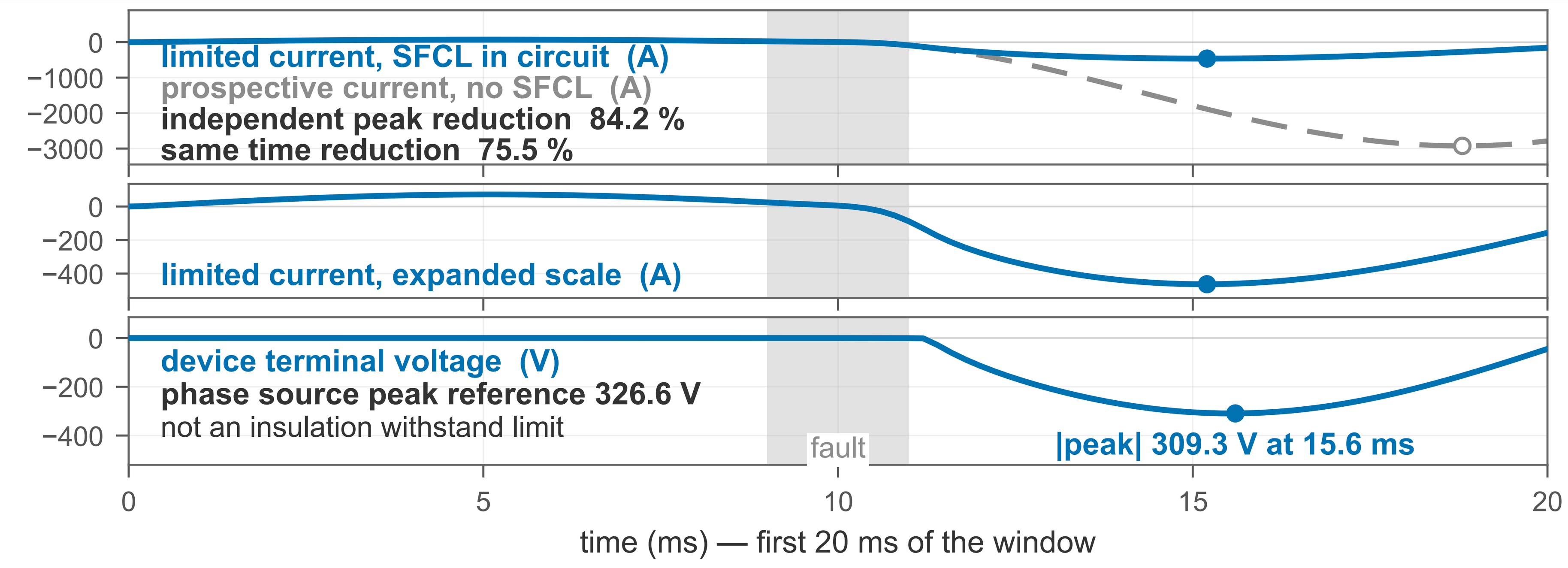




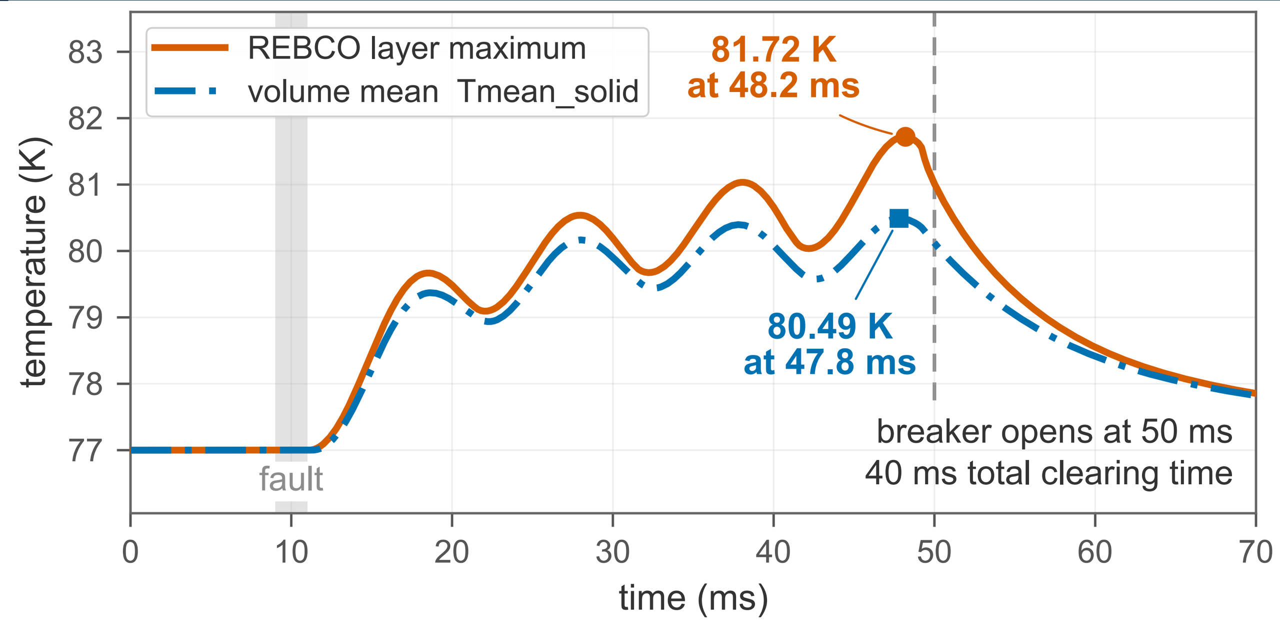


Two hairpin strings in parallel; in each, outgoing and return tapes are in series so their currents cancel. Panel **(b)** is the four stacked tapes, panel **(c)** the modelled segment and its length scaling.

2 Circuit and thermal response



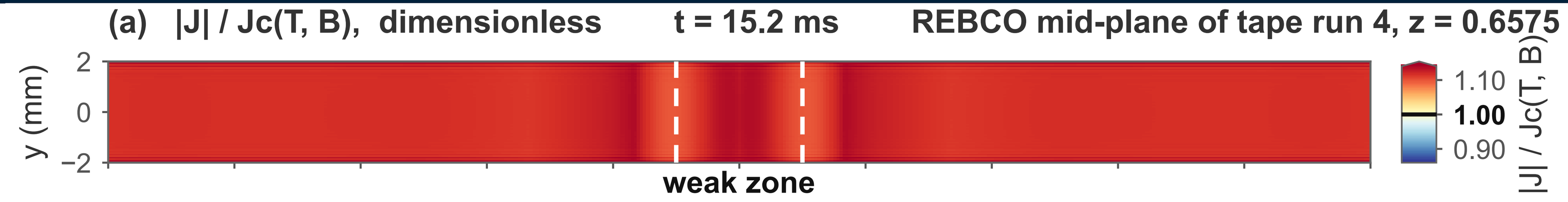
Current limiting and device terminal voltage on one shared time axis. The voltage peak is 0.4 ms after the current peak, so the two are not one operating point; the modelled peaks are negative and the sign carries no design meaning. The limited and prospective current peaks are 3.6 ms apart, so the independent and the same time reductions are different ratios.



Temperature, double reported, over the 40 ms clearing event: the volume mean of the 24 solid domains, and the 1 µm REBCO layer extremum that is never a design point value on its own. They peak 0.4 ms apart.

3 Spatial fields, model credibility and limitations

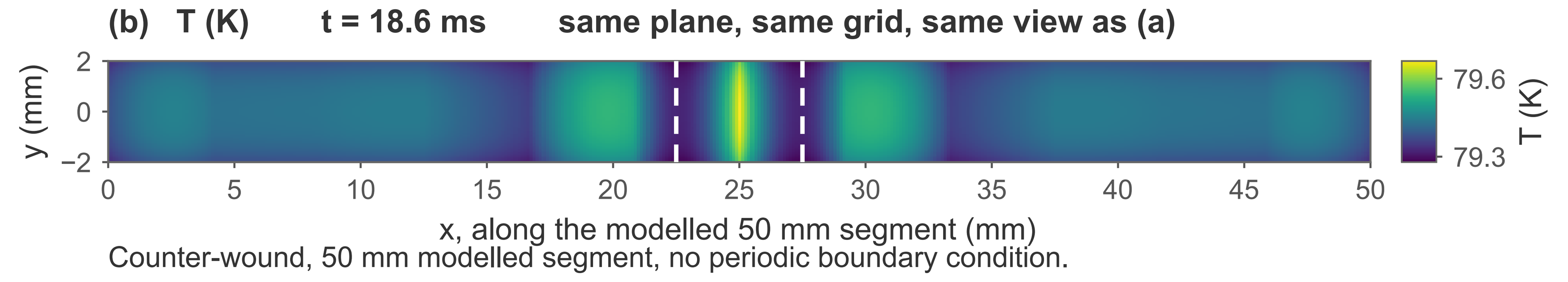
(a) |J| / Jc(T, B), dimensionless t = 15.2 ms REBCO mid-plane of tape run 4, z = 0.6575 mm



|J| is a curl derived field; the tape edge band and the position of its maximum are both set by the mesh.

the whole layer is above unity at this instant: 100.00 % of the sampled plane
weak zone: centre x = 25 mm, wx = 2.5 mm, minimum Jc/Jc0 = 0.85
(a) plane maximum is a sampled lower bound on the extremum over the domain
plane max 1.2309
on a tape edge row, |y| = 2.00 mm; x position set by the mesh

(b) T (K) t = 18.6 ms same plane, same grid, same view as (a)

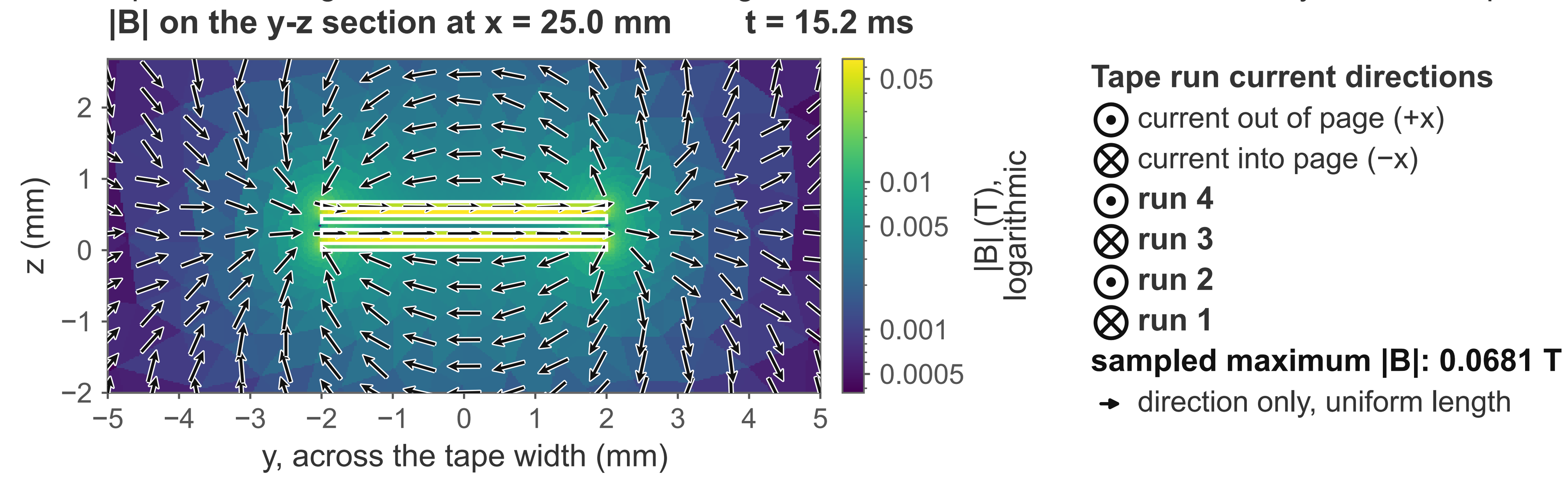


Counter-wound, 50 mm modelled segment, no periodic boundary condition.

plane max (grid sampled) 79.66796806 K · mean 79.41655240 K
min 79.27894020 K
Reference mesh field; hotspot magnitude remains mesh sensitive.
plane maximum at x = 25.00 mm, y = 0.00 mm

The same plane, same grid, at the two instants the figure names: normalised current density above, temperature below. Dashed lines mark the weak zone, and both panels share one axial axis.

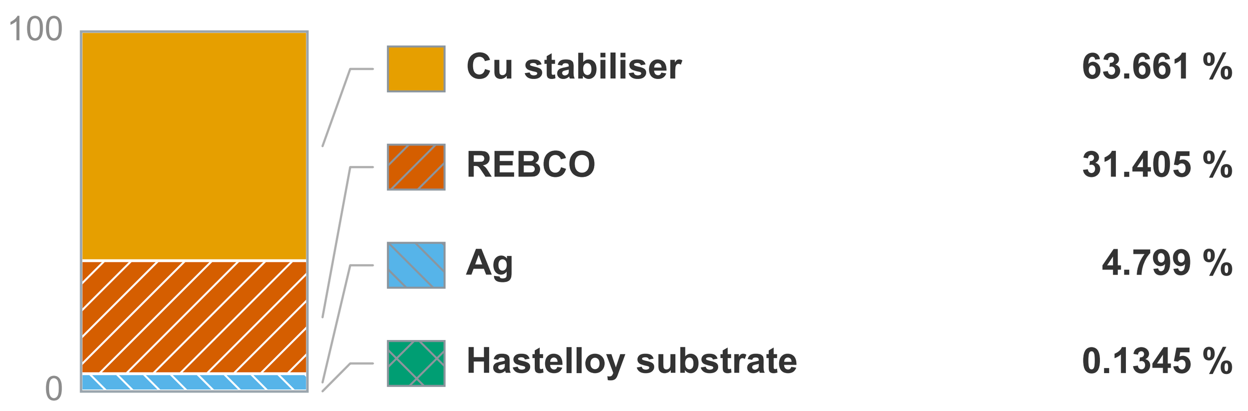
|B| on the y-z section at x = 25.0 mm t = 15.2 ms



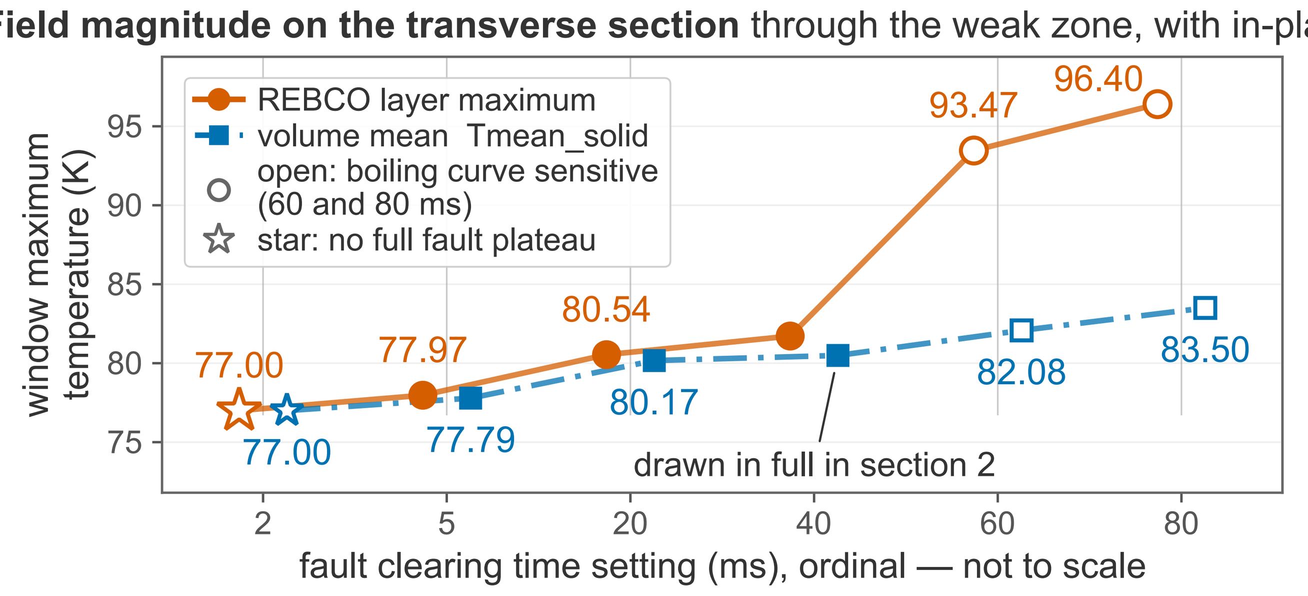
Direction symbols read from the exported field, by Ampère circulation around each tape run.

Field magnitude on the transverse section through the weak zone, with in-plane direction arrows.

share of window E·J energy (%)



Share of the modelled solid E · J energy over the window, by material group. It is a segment-level time integral, not an instantaneous power split, and length scaling does not change it.



Six discrete breaker settings, one run each. The calibres differ between settings, so the six are not one homogeneous scan, nothing between them is interpolated, and the two longest are sensitive to the boiling curve.

Model scope.
- Coupled 3D H-formulation, solid heat transfer and circuit in one unsegmented 70 ms window: 10 ms energising, 40 ms faulted, then 20 ms after the breaker opens.
- Two parallel strings, counter-wound to suppress net inductance; no external bypass.
- A 50 mm representative segment containing four tape runs, discretised with 77,928 elements at a relative tolerance of 10^{-4}; device values follow by length scaling.

Outstanding limitations.
- Axial mesh refinement has not entered an asymptotic range, and the thickness and transverse directions were never solved, so no error bar is quoted.
- The model is resolved only up to a 40 ms clearing time; the 60 ms and 80 ms settings remain unresolved, being sensitive to the boiling curve superheat axis, and typical switchgear is slower still.
- The 0.1 mm inter-tape spacing is assumed, as is one weak zone per 50 mm segment.
- Nothing here has been calibrated against an independent formulation.

Take home. Within this model line the device holds the line current far below the prospective fault current, the solid peaks under five kelvin above the bath, and most of the window E · J energy goes to the copper stabiliser.

References
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